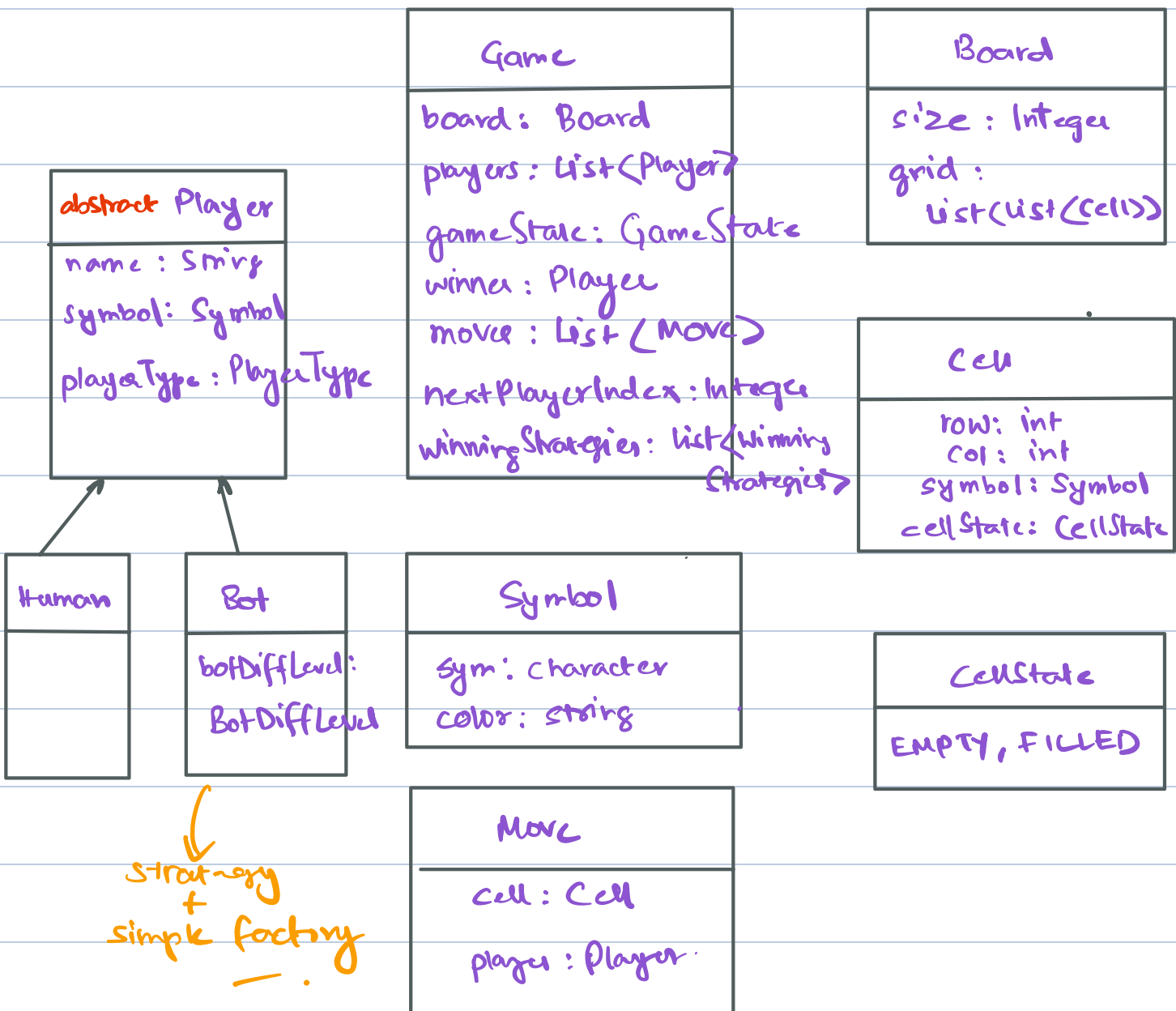


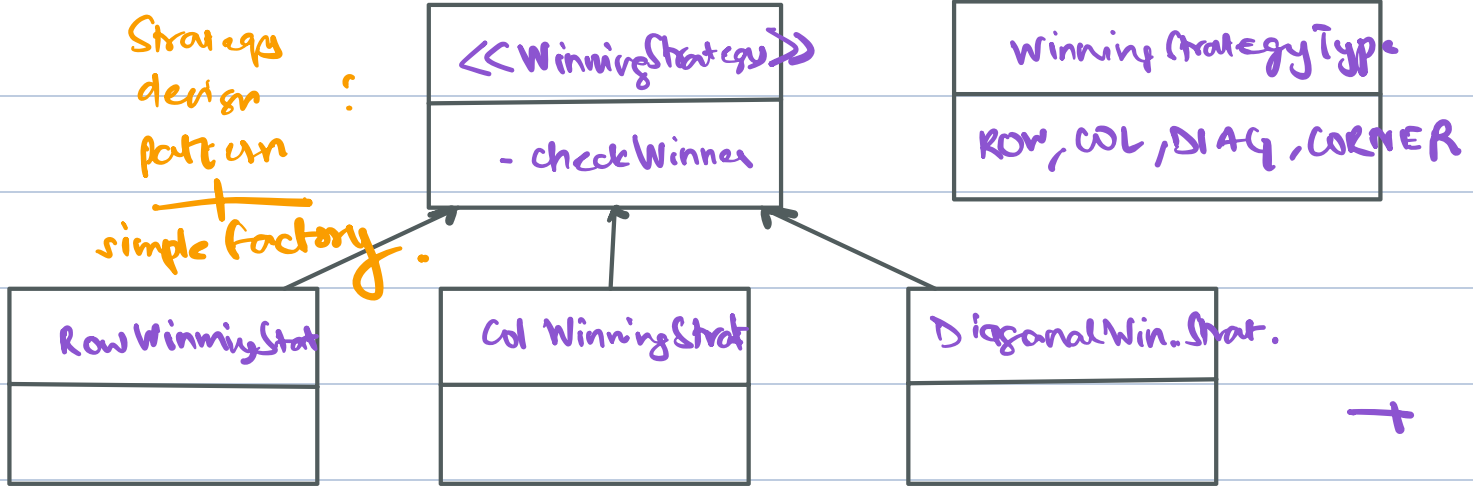
Agenda:

1. class Diagram
2. Check Winner's algorithm
3. coding (entities , controller , displaying the board)

Class Diagram:

Builder
design pattern .





Check Winner Algorithm:

Brute force solution:

for every player (1-N):

X		O
X	X	X
	O	O

// Row

for (i=0; i <= n; i++) {

is_winner = true;

for (j=0; j <= n; j++) {

if (mat[i][j].sym != player.sym)

{

is_winner = false;

break;

}

if (is_winner) {

return player;

Time complexity : $O(N^3)$

Solution 1:

checkWinner (Board, player) {

// Row

for(i=0; i <= n; i++) { → N

is_winner = true;

for(j=0; j <= n; j++) { → N

if (mat[i][j].sym != player.sym)

{ is_winner = false;

break;

}

if (is_winner) {

return player;

}

}

Time complexity : $O(N^2)$

Solution 2:

checkWinner (Board, Player, Cell) {

is_winner = true;

for (j = 0; j <= n; j++) {

if (mat[cell.row][j].sym != player.sym)

{
is_winner = false;
break;

}

if (is_winner) {

return player;

}

}

Time complexity : $O(N)$:

x		o
x	x	x
	o	o

Home work: Time complexity $O(1)$

Break till : 10:10 PM

Solution 3 :

$$\left\{ 0 \rightarrow \begin{Bmatrix} x:1 \\ o:1 \end{Bmatrix}, 1 \rightarrow \begin{Bmatrix} x:1 \\ o:2 \end{Bmatrix}, 2 \rightarrow \begin{Bmatrix} x:1 \text{ or } 3 \\ o: \end{Bmatrix} \right\}$$

$$\left\{ \begin{array}{l} 0 \rightarrow \begin{Bmatrix} x:1 \\ o:1 \end{Bmatrix} \\ 1 \rightarrow \begin{Bmatrix} x:2 \\ o:1 \end{Bmatrix} \\ 2 \rightarrow \begin{Bmatrix} x:1 \\ o:1 \end{Bmatrix} \end{array} \right\}$$

	0	1	2
0		0	x
1	0	x	x
2		0	x

HashMap < Integer, HashMap < Char, Int > >